Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

Holdwise

Project / Solution Name

Holdwise: WhatsApp enquiry-to-quotation module for Orbyte

Name (Leader)

Yerassyl Mubarak

Email & Phone (Leader)

ait2609221@xmu.edu.my / +77067374265

University Affiliation(s)

Xiamen University Malaysia (XMUM)

Live Prototype URL

https://quote-lah-hardware-quotation-mvp.replit.app

Code Repository URL (public / access granted to judges)

https://github.com/aithubxmu/Holdwise

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

A shop's WhatsApp enquiries (synthetic, simulated) appear in the Inbox of a Next.js 15 web app, exported as a static site and served by Replit Deployments. The browser calls an Express 5 REST API (Node.js, TypeScript) through typed TanStack Query clients generated from an OpenAPI spec.

When staff draft a quote, the API sends the message text and the catalogue's product names and aliases, never prices, to GLM-5.3 on the RunPod Public Endpoint (OpenAI-compatible). The reply must be strict JSON that passes a Zod schema; if the AI is off, slower than 8 seconds or invalid, a TypeScript rules parser is used instead.

A deterministic rules engine matches items to the catalogue, prices them, applies discounts and delivery fees, checks stock, raises review flags and routes large quotes to the Owner. Quotes, stock reservations, settings, credits and the activity log are stored in PostgreSQL through Drizzle ORM. Approved quotes go to a simulated WhatsApp outbox, and the dashboard reads only from the database.

Secrets live in Replit Secrets. The Orbyte hub is simulated (module shell and credit wallet); there is no real Orbyte or WhatsApp integration.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9; Node.js 24	Main language; server runtime	Apache-2.0; MIT
Next.js 15.5; React 19.1	Web app (App Router, static export) and UI	MIT
Express 5.3; pino 9	REST API server and request logging	MIT
Drizzle ORM 0.45; node-postgres 8.23	Database schema and queries for PostgreSQL	Apache-2.0; MIT
Zod 3.25; Orval 8; TanStack Query 5.90	Validation of API input and AI JSON; generated typed API clients	MIT
Tailwind CSS 4.1; Radix UI; lucide-react 0.545	Styling, accessible UI components, icons	MIT; MIT; ISC

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
GLM-5.3 Public Endpoint (RunPod; model by Z.ai)	The only AI step: reads each WhatsApp message into JSON (items, quantities, units, confidence, delivery area)	Sponsor-provided (RunPod US$25 team credit)
Google Gemini API (gemini-3.5-flash, flash-lite)	Optional backup extraction provider, used during development; switched by environment variable, not active in the deployed app	Free tier
None other	WhatsApp sending is simulated; no messaging, maps or payment APIs	Not applicable

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments (Replit)	Hosts the live web app and API at quote-lah-hardware-quotation-mvp.replit.app; Replit Agent helped build the first version	Replit Core (sponsor voucher); Autoscale 2 vCPU / 4 GiB, Asia
Replit PostgreSQL (Replit)	Stores catalogue, customers, quotes, stock reservations, settings, credits and the activity log	Included with Replit; Replit-managed
Replit Secrets (Replit)	Holds RUNPOD_API_KEY and other keys; never in code	Included with Replit
RunPod Serverless Public Endpoint (RunPod)	Runs GLM-5.3 inference for message extraction	Pay per token from sponsor credit; RunPod-managed
GitHub (GitHub)	Public code repository; the Repl pulls main and is republished	Free

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
GLM-5.3 (zai-org/ glm-5.3)	Z.ai, via RunPod Hub Public Endpoint	Pre-trained; prompted to return strict JSON, no fine-tuning	RunPod Terms of Service; Z.ai model terms
Gemini 3.5 Flash / Flash-Lite (optional backup)	Google, Gemini API	Pre-trained; same prompt; not active in the deployed app	Gemini API Terms of Service
No other models	None	No fine-tuned or self-trained models	Not applicable

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Demo catalogue (40 products)	Repo: lib/db/ src/demo	Synthetic, self-made: names, aliases, prices, stock	Apache-2.0 (repo)	N
Demo customers, zones, users	Repo: lib/db/ src/demo	Synthetic: fictional "(demo)" names, placeholder phones	Apache-2.0 (repo)	N
Demo WhatsApp messages (15)	Repo: lib/db/ src/demo	Synthetic: Malay, English, Manglish	Apache-2.0 (repo)	N
AI evaluation set (20 messages)	Repo: data/ eval-set.json	Synthetic: 15 seed + 5 held-out, expected SKU and qty	Apache-2.0 (repo)	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude Code (Anthropic), Claude Opus 5.5	Writing and refactoring code (API, rules engine, AI adapter, UI), tests, documentation and debugging the deployment	Major
Replit Agent (Replit)	First scaffold of the monorepo, inbox screen and Replit deployment setup	Moderate
Claude (Anthropic), chat assistant	Drafting the pitch deck, PRD and declaration text	Moderate

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-written changes were reviewed by the team before merging through GitHub pull requests. We ran the unit tests (31 tests, including the Zod schema, the AI fallback and a check that AI cannot change prices), an end-to-end script of 39 API checks, and clicked through the full chain in the browser with AI on and with no key. A 20-message evaluation set measures the product's AI against the rules parser. AI never sets prices, uncertain lines are flagged, and every quote needs human approval.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod	GenAI API and credits: GLM-5.3 Public Endpoint, US$25 team credit	Main AI model that reads WhatsApp messages into JSON for every drafted quote
Replit	Platform: Replit Core (voucher), Replit Agent, Deployments, PostgreSQL, Secrets	Development, hosting of the live prototype, database and secret storage

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to how the prototype is delivered: it hosts the live app, database and secrets. RunPod's GLM-5.3 powers the AI step but is replaceable by design: a thin adapter switches providers through environment variables (for example Gemini), and if no model is reachable the rules parser keeps the whole quote pipeline working, with lower accuracy on unseen messages (5/8 items instead of 8/8 on our held-out set). The sponsor AI improves quality but is not a single point of failure.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

No real user data is collected. The prototype uses synthetic data only: fictional customers marked "(demo)", placeholder phone numbers and invented messages. There is no real sign-in (roles are simulated) and no real WhatsApp connection.

When AI extraction is on, the text of a demo message and the product names are sent to the GLM-5.3 endpoint on RunPod; customer names, phone numbers and prices are not included. Data is stored in Replit-managed PostgreSQL, and the owner can wipe and re-seed it with "Reset demo data". API keys are kept in Replit Secrets and never committed.

For a real deployment in Phase 2 we would follow Malaysia's PDPA 2010: a consent notice, data minimisation (removing names and phone numbers before any AI call), a fixed retention period for messages, and access limited by role.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination: the model could invent items or quantities. It never sees or returns prices, its output must pass a strict Zod schema, items are matched only to the catalogue, unknown ones are flagged, and a variant the customer did not name is discarded.

Uncertainty: low confidence (below 0.7), missing quantities, ambiguity and low stock are flagged and must be resolved by a person; every quote needs approval, and large ones need the Owner.

Single provider and rate limits: an 8-second timeout and a rules fallback, with the reason shown on the draft; providers switch by environment variable.

Limited evidence: our evaluation set is small (20 synthetic messages) and is not a benchmark; Phase 2 needs real, anonymised messages. Only Malay, English and Manglish are tested.

Cost: at most 1,000 output tokens per call with low reasoning effort.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



Yerassyl Mubarak

[Date]

24 October 2026